



ICT 21633

Fundamentals of Software Engineering

Assignment - 02

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2000 Level

1st Semester

Department of Information
Technology

University of Ruhuna

Introduction

SDLC known as software development life cycle. It is a structured process used to planning, creating, testing and deploying software systems. There are some main phases of the software development life cycle. They are, Planning, Analysis, Design, Implementation, Testing and integration, deploying, Maintenance. Each phase is crucial for creating a successful software product. Furthermore each phase involves different activities. Using SDLC we can design and build high quality software. It is a cost effective and time efficient process.

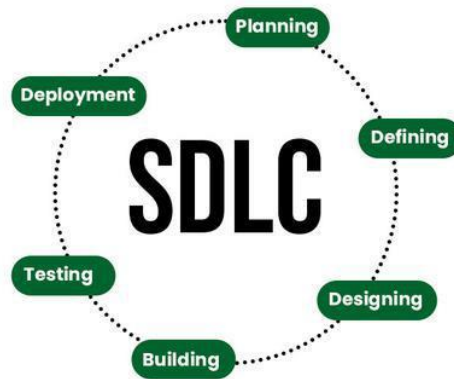


Figure 01-Software Development Life Cycle

Planning Phase

In the planning phase of the software development life cycle objectives, project's scope and feasibilities are defined. In this phase stakeholders identify goals, resources and timelines of the project. Furthermore stakeholders understand client or end user needs. In this phase stakeholders assess the technical, operational and financial feasibilities of project. Furthermore stakeholders have to define what should be included and excluded in the project. Determining required resources such as human, technological and financial. Furthermore stakeholders have to create a project schedule in this phase.

Analysis Phase

In this phase analysis refers to the process of understanding and specifying what the software needs to do to meet user requirements. This is the process of gathering information about the needs and expectations of stakeholders for the software system. This step involves interviews(unstructured), surveys, focus groups, workshops, mind mapping and other techniques to gather information from stakeholders. After that analyzing the information, identify the high level goals and objectives of the software system. Furthermore identifying any constraints or limitations that may affect the development of the software system. Moreover in this phase stakeholders create a clear documentation and check that the requirements are complete and accurate.

Design phase

System's architecture and detailed specifications are created in design phase using requirements gathered in the analysis phase. Describes the overall structure of the system including the choice of technology stack, system components and their interaction. Stakeholders create database schema in good manner to store data efficiently and design user friendly user interface. Components of the system break down to smaller modules to define their functionalities and interfaces. The effective design ensures that the development process is efficient and risk are minimized. Furthermore development group create detailed specifications for each component.

Implementation phase

In software development life cycle actual coding of the software is included in implementation phase. In this phase programmers translate design documents and specifications into software code using programming languages and tools. Furthermore programmers will check individual modules of the software system to ensure they work correctly. After that programmers combine individual modules and test them together. The main goal of the implementation is produce a good software product that meets client's requirements. There are lots of programming languages in today's world. Some of them are, Java, Python, Java Script, C#, PHP.

Testing and Integration phase

In testing and integration of SDLC Testers and Quality Assurance engineers ensure that software functions correct and meets user requirements. In this phase Testers identify bugs in the software system. There are four types of testing in this phase. They are, Unit, Integration, System and Acceptance test. Unit testing means Testers or QA engineers test individual components and functions in isolation. Integration testing means testers or QA engineers test interactions between integrated units or modules. In system testing they evaluates the complete and integrated software system. In acceptance testing they validates the software for user requirements and business needs.

In integration phase Testers or QA engineers combine and test various modules of the software to ensure that they work together as intended. In this phase tester use automated builds and tests to detect integration issues early. Using testing and integration testers and QA engineers ensure that each component is working correct within the system.

Deployment phase

In deployment phase end user or client receive the completed software. Client will ensure that the development environment is ready including hardware, software and network configurations. After that client will install the software and adjust settings to fit the specific needs. Furthermore client will check the software to confirm that the software works as expected. After that client will publish the software to user. Furthermore client will monitor the software's performance and functionality to find any issues available in software.

Maintenance phase

In maintenance phase software is updated, repaired and improved after the software initial release. In this phase system administrators can correct errors that they could not find errors during the initial phases. Also they can add new features or they can improve existing ones. Furthermore they can improve software efficiency and performance in this phase. Also system administrators can apply updates for hardware, software and security environments. The goal of maintenance is to ensure that the software continues to meet user needs and operate effectively.

Summary With a Example

Let's think that our development team is going to do a Hotel Reservations System software, then our group have to planning about the system. In this phase our development team will ask about requirements of the client. After that we have to identify project goals, resources and timelines of the project. In this phase we have to assess feasibilities such as technical, operational and financial. After that we have to create a project schedule including deadlines and timelines.

In analysis phase we have to gather information about the project needs and expectations of our working group using such as interviews, surveys and focus groups. We have to conduct interviews with hotel staff, customers and management to gather requirements. In hotel reservations systems, key requirements are,

- Hotel package details
- Payment process
- Book or reserve rooms
- Facilities
- Check in and outs
- Security

- Location/ Hotel history
- Staff and service details

Our development group have to gather functional requirements like above. Furthermore we have to consider about non functional requirements like system performance, security. We can use diagrams such as Data flow diagram, HIPO diagram, UML diagram for both analysis and design phase.

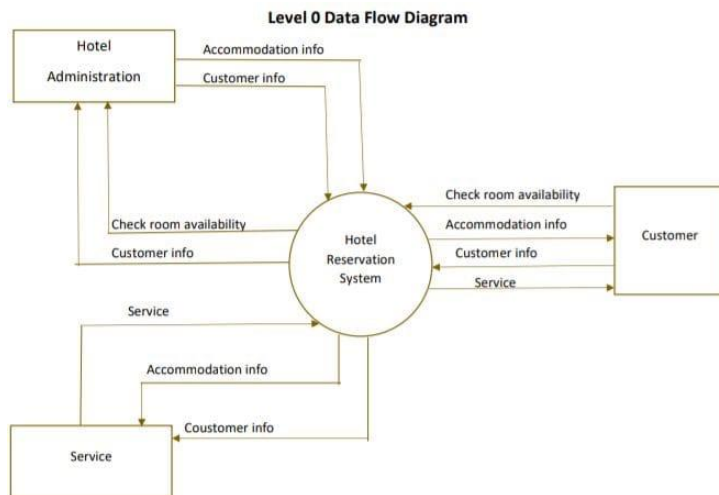


Figure 02-Level 0 Data Flow Diagram

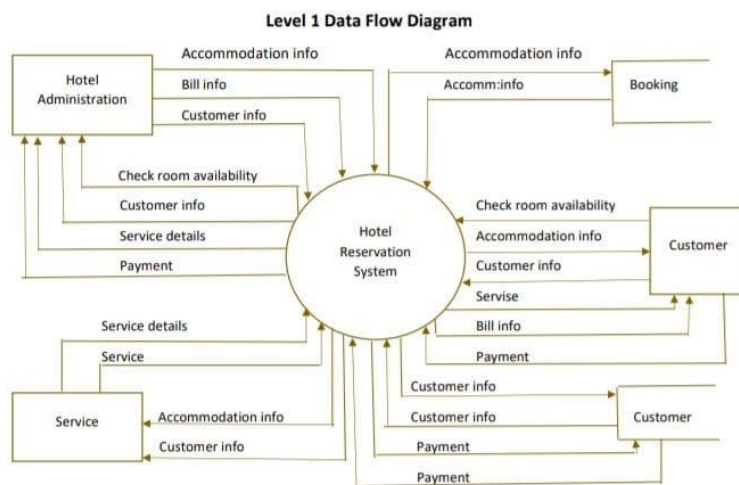


Figure 03-Level 1 Data Flow Diagram

In the design phase development group have to create blueprint for the system based on the requirements. Like that, we have to create development system architecture, database schema and user interface design.

Furthermore programmers have to coding the system using coding languages. After that programmers have to implement database integration and they have to ensure that all the components work together. Our testers have to check the system including such as unit testing, integration testing, system and user acceptance testing. After those process our development team can release the system for hotel and it's customers .After that hotel administration check and use the hotel reservation system. Finally system administrators update, repair and improve the hotel reservation system.

According to above phases, software development life cycle is very useful for stakeholders to create a high quality software efficiently and systematically. We can use software models for software development such as Waterfall, Agile, Iterative and Spiral model. Those models were introduced to follow a disciplined and systematic method while designing software.

References-

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